RecruitShield — AI + Blockchain Powered Recruitment Scam Detection System

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1. Problem Understanding

Recruitment fraud has evolved into an organized, multi-channel cybercrime industry — no longer just "obvious" scam emails, but sophisticated operations using cloned company domains, deepfake HR profiles, and spoofed offer letters that pass casual visual inspection.

The core failure point: trust verification in hiring is currently **manual, fragmented, and non-standardized**. A candidate has no single source of truth to verify:

- Is this recruiter actually employed by the company they claim?
- Has this exact posting or message pattern been reported as fraudulent before?
- Is money ever supposed to flow from candidate → employer? (Answer: never — yet millions don't know this.)

Why existing solutions fail:

Existing Approach	Limitation
LinkedIn/Naukri manual reporting	Reactive, slow, siloed per platform — a scam banned on one platform reappears on WhatsApp/Telegram instantly
Company HR verification hotlines	Doesn't scale, most small companies don't have one
Manual WHOIS/Google searching by candidates	Requires cybersecurity literacy most job seekers don't have
Antivirus/browser phishing filters	Detect malicious links, not socially-engineered *text-based* scams (no malware payload = invisible to them)

The real gap: There is no **decentralized, tamper-proof, cross-platform trust layer** for recruitment — every platform re-invents scam detection in its own silo, and scammers exploit exactly that fragmentation by hopping platforms the moment they're flagged.

This is fundamentally a **trust and identity verification problem wearing an HR costume** — which is precisely why it's solvable with the same primitives cybersecurity and blockchain systems already use for identity and tamper-evidence.

Recruitment scams exploit fragmented trust systems, migrating seamlessly across platforms and bypassing traditional defenses. Victims face identity theft, financial loss, and eroded confidence in hiring. Without a decentralized, tamper-proof verification layer, detection remains reactive and siloed, leaving millions vulnerable to increasingly sophisticated, cross-channel fraud operations.

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2. Proposed Solution

RecruitShield is a **multi-layered Trust Verification Engine** that scores any job posting, recruiter message, or offer letter in real time — combining NLP-based scam-pattern detection, cybersecurity-grade identity verification, and a **blockchain-anchored, tamper-proof scam registry** that no single platform or bad actor can manipulate.

The Core Insight (your differentiator)

Instead of one AI model guessing "real or fake," RecruitShield runs **four independent verification layers in parallel**, and — critically — writes every confirmed scam report to an **immutable, append-only blockchain ledger**. This means:

- No scammer can bribe, hack, or pressure a moderator to delete their scam record (unlike centralized report databases)
- Every flagged scam has a **public, auditable trail** — verifiable by anyone, including researchers, journalists, and law enforcement
- The registry becomes a **shared public good** across platforms — instead of every job portal rebuilding scam detection from scratch, they can all query the same trust layer via API

Why it is unique?

Anyone can build a scam-classifier model. What's much harder to replicate is a **cross-platform, tamper-proof source of truth** that scammers cannot erase or manipulate once flagged — that's the blockchain angle earning its place, not bolted on for buzzword points.

In future, RecruitShield extends beyond detection by offering a centralized web dashboard, multilingual scam analysis, and API integrations for job portals. Community-powered reporting strengthens the blockchain registry, while law enforcement and researchers gain auditable access. This ecosystem transforms RecruitShield into a scalable, transparent trust layer for global recruitment security.

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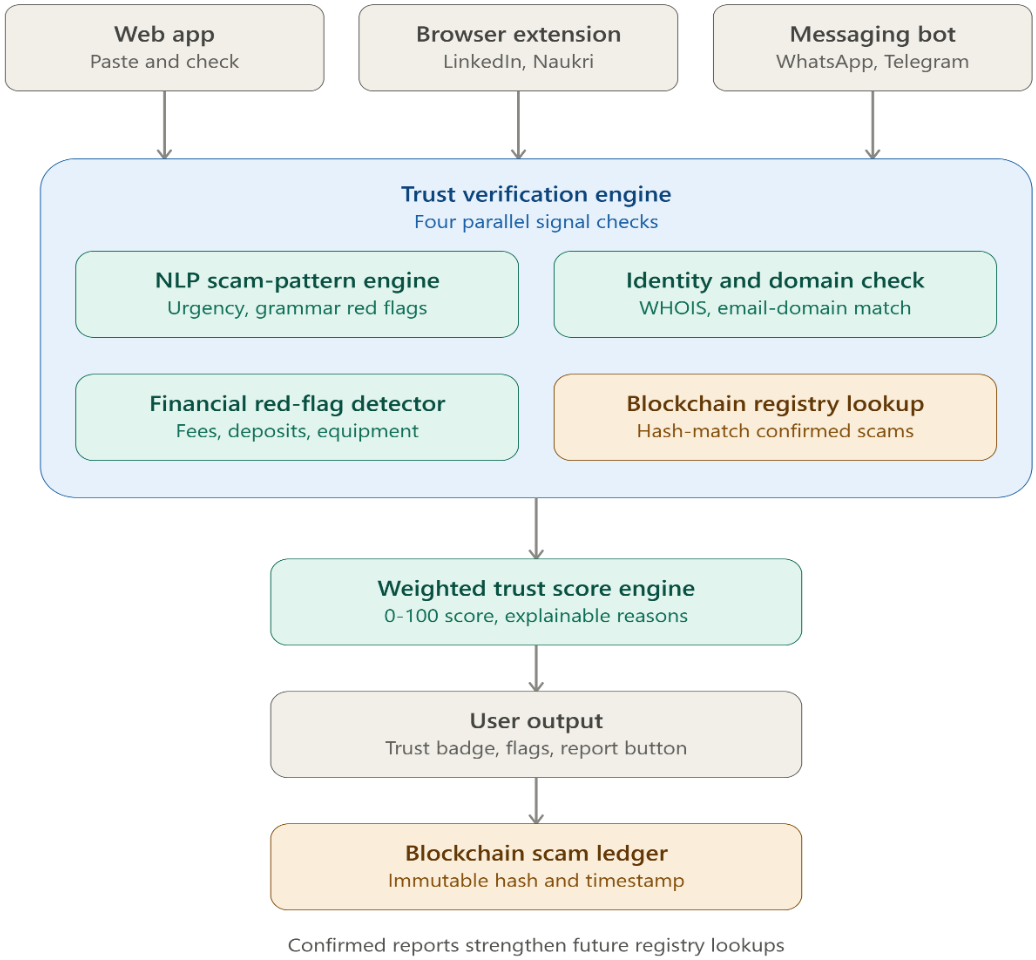
3. System Architecture



How the blockchain piece works (kept simple, not over-engineered):

- We don't put raw personal data on-chain (expensive, privacy-violating, and unnecessary).
- We hash the key scam-identifying fields (recruiter email domain, company name claimed, phone number, key phrases) using SHA-256, and store **only the hash + a risk category + timestamp** on a lightweight blockchain (e.g., Polygon testnet or Hyperledger for hackathon scope).
- When a new posting comes in, we hash it the same way and check for a match against the on-chain registry — fast, private, and tamper-proof.
- This is the same pattern used in real-world breach-checking systems like "Have I Been Pwned" — proven, explainable, and easy for judges to verify live.

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4. Key Features

1. **Real-Time Trust Score (0–100)** — instant, color-coded verdict (🟢/🟡/🔴) for any pasted job posting, email, or message.
2. **Explainable Red-Flag Breakdown** — not a black box; shows exactly *why* something is risky (e.g., "⚠️ Recruiter domain registered 9 days ago," "⚠️ Requests refundable security deposit").
3. **Blockchain-Anchored Scam Registry** — every confirmed scam report is hashed and written to an immutable ledger, making the data tamper-proof and independently auditable — a first-of-its-kind trust layer for recruitment fraud.
4. **Cross-Platform Browser Extension** — flags risky postings directly inside LinkedIn, Naukri, and Indeed while the user is browsing — no copy-pasting required.
5. **WhatsApp/Telegram Bot Integration** — since most real-world scams happen off-platform in chat apps, users can forward a suspicious message to a bot and get an instant verdict.
6. **Company & Recruiter Identity Verification** — cross-checks recruiter email domain against the claimed company's verified domain, using WHOIS + optional LinkedIn/company page validation.
7. **Community-Powered Crowdsourcing** — user-submitted scam reports strengthen the registry for everyone, with abuse-resistance via the blockchain layer (reports can't be silently deleted or manipulated).
8. **Offer Letter / PDF Authenticity Check** *(stretch feature)* — basic metadata and formatting anomaly detection to flag doctored offer letter PDFs.

In future we can also integrate

1. **Web Dashboard Analytics —** a centralized hub showing scam heatmaps, recruiter trust scores, and fraud trend reports for transparency and judge appeal.
2. **Multilingual Scam Detection —** support for regional languages (Tamil, Hindi, etc.) to expand accessibility and impact in India and beyond.
3. **Predictive Fraud Analytics —** AI forecasts scam spikes by platform, region, or time, helping job portals prepare proactively
4. **Law Enforcement API Access** — secure endpoints for investigators and researchers to query scam records, boosting credibility and social impact.
5. **Automated Scam Pattern Updates —** weekly refresh of red-flag rules and NLP models to catch emerging tactics like new messaging apps or payment schemes**.**

5. Technology Stack

Layer	Technology	Why
Frontend	React.js + Tailwind CSS	Fast to build, clean UI for demo
Browser Extension	JavaScript (Chrome Extension Manifest V3)	Direct in-platform flagging
Backend/API	Node.js (Express) or Python (FastAPI)	Lightweight, fast to prototype
NLP/ML Model	Scikit-learn (Logistic Regression/SVM) or DistilBERT (Hugging Face, free)	Trained on public "Fake Job Postings" Kaggle dataset — free, no GPU cost required
Identity Verification	WHOIS API (free tier), regex-based email-domain matcher	No-cost, high-precision checks
Blockchain Layer	Solidity smart contract on Polygon Mumbai Testnet (or Hyperledger Fabric for enterprise framing)	Free testnet gas, fast to deploy, widely understood by judges
Database (off-chain)	MongoDB / Firebase	Stores non-sensitive metadata, UI-facing data
Bot Integration	WhatsApp Business API (sandbox) / Telegram Bot API (free)	Reaches scams where they actually happen
Hosting	Vercel/Netlify (frontend), Render/Railway (backend)	Free-tier hackathon-friendly deployment

Note for judges: Every component above has a free tier sufficient for hackathon demo and even early pilot — this is a deliberate design choice, not a limitation

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6. Implementation Plan

Phase 1 — Core Detection Engine

- Collect/prepare labeled dataset (Kaggle "Real or Fake Job Postings" dataset)
- Train baseline NLP classifier (Logistic Regression → upgrade to DistilBERT if time permits)
- Build rule-based financial red-flag detector (regex layer)

Phase 2 — Identity Verification Layer

- Integrate WHOIS API for domain-age lookup
- Build email-domain vs. company-name matching logic
- Combine Layer 1 + Layer 2 + Layer 3 into weighted Trust Score function

Phase 3 — Blockchain Registry

- Write and deploy simple Solidity smart contract (store hash, risk category, timestamp)
- Connect backend to smart contract via Web3.js/Ethers.js
- Implement "Report Scam" → hash → write-to-chain flow

Phase 4 — Frontend + Integration

- Build web app UI (paste-and-check flow with Trust Score visualization)
- Build Chrome extension MVP for LinkedIn flagging
- Connect WhatsApp/Telegram bot sandbox for message-forwarding flow

Phase 5 — Polish, Testing & Pitch Prep

- End-to-end testing with real and synthetic scam examples
- Prepare live demo script (paste a known scam posting → show full breakdown → show blockchain record)
- Build pitch deck emphasizing the **tamper-proof registry** as the unique moat

Post-Hackathon Roadmap:

- Partner with job portals via API for platform-wide integration
- Expand blockchain registry into a public, queryable API (like a "VirusTotal for job postings")
- Add multilingual support for regional-language scam detection (high impact in India)

"RecruitShield is a tamper-proof, explainable trust layer for hiring — combining AI-driven scam detection with a blockchain-anchored registry that no scammer, platform, or bad actor can manipulate or erase."